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AEROSOL-GENERATING SYSTEM

Abstract

There is provided an aerosol-generating system, for providing an aerosol for inhalation by a user, comprising: an aerosol provision device comprising control circuitry for controlling an activation state of the aerosol provision device: a detector arranged to detect a first property associated with aerosol generating material, in use arranged in the aerosol provision device, and provide a signal to the control circuitry; and, age verification means for verifying an age of a user, wherein the control circuitry is arranged to perform at least one of i) update an activation state of the aerosol provision device, and ii) activate the age verification means in response to receiving a signal from the detector.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to an aerosol-generating system, an aerosol provision device, a method of providing an aerosol for inhalation by a user, and aerosol-generating means.

BACKGROUND

[0002] Aerosol-generating systems are known. Common systems use heaters which are activated by a user to create an aerosol by an aerosol provision device from an aerosol generating material which is then inhaled by the user. The device may be activated by a user at the push of a button or merely by the act of inhalation. Modern systems can use consumable elements containing the aerosol generating material. It can be desirable for the manufacturer to enable control over the activation of the systems. This may avoid the activation of the system in undesirable circumstances.

[0003] The present invention is directed toward solving some of the above problems.

SUMMARY

[0004] Aspects of the invention are defined in the accompanying claims.

[0005] In accordance with some embodiments described herein, there is provided an aerosol-generating system, for providing an aerosol for inhalation by a user, comprising: an aerosol provision device comprising control circuitry for controlling an activation state of the aerosol provision device; a detector arranged to detect a first property associated with aerosol generating material, in use arranged in the aerosol provision device, and provide a signal to the control circuitry; and, age verification means for verifying an age of a user, wherein the control circuitry is arranged to perform at least one of i) update an activation state of the aerosol provision device, and ii) activate the age verification means in response to receiving a signal from the detector.

[0006] Such a system is able to identify the aerosol generating material intended for use with the aerosol provision device, ascertaining if it is an age-restricted aerosol generating material and then updating an activation state of the aerosol provision device and/or activate an age verification means. This allows the system to determine if age verification is required (as a result of the aerosol generating material intended for use) and then perform age verification if it is. This provides an increased in-use safety aspect for the system while avoiding unnecessary interference in circumstances that do not require age verification. This therefore strikes a balance between overly strict and overly lenient control over the usage of the aerosol provision device.

[0007] In particular, in the event that consumable **1** is supplied for use with the aerosol provision device, consumable **1** containing an age-restricted aerosol generating material, age verification can be requested prior to operation of the device. In contrast, when consumable **2** is supplied for use with the aerosol provision device, consumable **2** not containing an age-restricted aerosol generating material, the device is allowed to operate without age verification. In this way, the manufacturer can provide a device that has a broad utility across a number of consumables and users, while protecting the device from being used with aerosol generating material that is unsuitable for a specific user.

[0008] The user experience of the device may also be increased by having a personalisable, “plug-and-play” approach. In that, the user may be provided with a number of options from which to select the operating state that the user desires. In this way, the user need not create, from scratch, an operating state for use with a consumable (containing a specific aerosol generating material), but can refine their preference from a number provided. In this way, the safety of the device is also maintained as each of the possible selectable operating states are known by the manufacturer to

provide a suitable aerosol without damaging the device. Therefore, both the device and user benefit from the arrangement. Furthermore, the user may only be able to access some of the operating states following age verification-increasing the overall safety of use of the device.
[0009] By requiring detection of a first property associated with aerosol generating material in-use combined with the aerosol provision device for providing an aerosol for inhalation by a user, the system may hinder use of non-authentic consumables. In particular, if the identification method is not compatible with the aerosol generating material (by virtue of it being non-authentic) the device may not be able to operate. In this way, the device can be protected from use with unknown and un-tested consumables (containing unknown and/or un-tested aerosol generating material) that might provide a substandard aerosol for the user or damage the device in an attempt to produce an aerosol.
[0010] The aerosol provision system of the present invention is able to operate in “offline” or “online” mode when performing age verification of potential users. In this way, a valid user may operate the system in offline (unconnected to a mobile network or internet or the like) environment provided the user satisfies the criteria for operation. The user experience of the device is thereby improved. This can be achieved by including a memory in the aerosol provision system with which the age verification means communicates.
[0011] In accordance with some embodiments described herein, there is provided an aerosol provision device, for providing an aerosol for inhalation by a user, comprising: control circuitry for controlling an activation state of the aerosol provision device; a detector arranged to detect a first property associated with aerosol generating material, in use arranged in the aerosol provision device, and provide a signal to the control circuitry; and, age verification means for verifying an age of a user, wherein the control circuitry is arranged to perform at least one of i) update an activation state of the aerosol provision device, and ii) activate the age verification means in response to receiving a signal from the detector.
[0012] In accordance with some embodiments described herein, there is provided a method of providing an aerosol for inhalation by a user, the method comprising: detecting, by a detector, a first property associated with aerosol generating material arranged in an aerosol provision device; providing, by the detector, a signal to control circuitry of the aerosol provision device; in response to receiving, at the control circuitry, a signal from the detector, performing at least one of: updating, by the control circuitry, an activation state of the aerosol provision device; and activating, by the control circuitry, age verification means.
[0013] In accordance with some embodiments described herein, there is provided aerosol-generating means, for providing an aerosol for inhalation by a user, comprising: an aerosol provision device comprising control means for controlling an activation state of the aerosol provision device; detecting means arranged to detect a first property associated with aerosol generating material, in use arranged in the aerosol provision device, and provide a signal to the control means; and, age verification means for verifying an age of a user, wherein the control means is arranged to perform at least one of i) update an activation state of the aerosol provision device, and ii) activate the age verification means in response to receiving a signal from the detecting means.

Description

DESCRIPTION OF DRAWINGS
[0014] The present teachings will now be described by way of example only with reference to the following figures:
[0015] FIG. **1** is a schematic view of an aerosol provision system according to an example;
[0016] FIG. **2** is a schematic view of an aerosol provision system according to an example;

[0017] FIG. **3** is a schematic view of an aerosol provision system according to an example;
[0018] FIG. **4** is a schematic view of an aerosol provision system according to an example; and,
[0019] FIG. **5** is a flow diagram according to an example.
[0020] While the invention is susceptible to various modifications and alternative forms, specific embodiments are shown by way of example in the drawings and are herein described in detail. It should be understood, however, that the drawings and detailed description of the specific embodiments are not intended to limit the invention to the particular forms disclosed. On the contrary, the invention covers all modifications, equivalents and alternatives falling within the scope of the present invention as defined by the appended claims.

DETAILED DESCRIPTION

[0021] Aspects and features of certain examples and embodiments are discussed/described herein. Some aspects and features of certain examples and embodiments may be implemented conventionally and these are not discussed/described in detail in the interests of brevity. It will thus be appreciated that aspects and features of apparatus and methods discussed herein which are not described in detail may be implemented in accordance with any conventional techniques for implementing such aspects and features.
[0022] The present disclosure relates to aerosol provision systems, which may also be referred to as aerosol provision systems, such as e-cigarettes. Throughout the following description the term “e-cigarette” or “electronic cigarette” may sometimes be used, but it will be appreciated this term may be used interchangeably with aerosol provision system/device and electronic aerosol provision system/device. Furthermore, and as is common in the technical field, the terms “aerosol” and “vapour”, and related terms such as “vaporise”, “volatilise” and “aerosolise”, may generally be used interchangeably.
[0023] FIG. **1** illustrates a schematic view of an example of an aerosol provision system **100** according to the present invention. The aerosol provision system **100** comprises an aerosol provision device **110**. The aerosol provision device **110** has control circuitry **120**. The control circuitry **120** is arranged to control an activation state of the aerosol provision device **110**. The aerosol provision system **100** comprises a detector **130** arranged to detect a first property associated with aerosol generating material, in-use arranged in the aerosol provision device **110** and provide a signal to the control circuitry **120**. The aerosol provision system **100** further comprises age verification means **140** for verifying an age of a user. The control circuitry **120** is arranged to perform at least one of update an activation state of the aerosol provision device **110** and activate the age verification means **140**.
[0024] In an example, the detector **130** detects a property of the aerosol generating material **132** to ascertain whether age verification is required prior to use of the aerosol generating material **132** in the aerosol provision device **110**. The property of the aerosol generating material **132** that may impact the requirement of age verification may be any of a chemical composition of the aerosol generating material, a strength of the aerosol generating material, a size of the consumable, a rating of the consumable, an authenticity of the consumable; amount; date of production; and, batch number or the like.
[0025] The detector **130** detects the property and provides a signal to the control circuitry **120**. The control circuitry **120** may process the signal and assess whether age verification is required. If age verification is required, the control circuitry **120** activates the age verification means **140**. The user is then informed that to activate the device **110**, age verification must be satisfied. Following age verification, the control circuitry **120** can assess whether the user is a valid user for use with the identified aerosol generating material **132** or not. If the user is valid, the device **110** is allowed to activate to provide an aerosol for inhalation to the valid user. If the user is not valid, the control circuitry **120** prevents activation of the device **110**.
[0026] In this way, age verification is only provided when required. In this way a balance is struck between impacting the user experience for safety and the safety of use of the device.

[0027] The full range of operating states of the aerosol provision system **100** may be very wide ranging such that the aerosol provision device **110** can be used with a correspondingly wide range of aerosol generating materials **132**. In this way, the user need not be limited to only certain types of aerosol generating material **132** for use with the aerosol provision device **110**. However, from within this wide range, some operating states will not be suitable for use with some aerosol generating materials **132**. As such, it is important to offer states that will provide a desirable aerosol from the aerosol generating material **132**. By offering all the possible states to all consumables, the onus would be put onto the user to work out which operating states interact well with which consumables. This may decrease the user experience of the system **100**.
[0028] The aerosol provision device **110** may comprise heating arrangements or the like for providing an aerosol from a consumable—the consumable may contain some aerosol generating material or the like. The control circuitry **120** may control the heating arrangement (or the like) according to the signal received from the detector **130**. The detector **130** enables the control circuitry **120** to ascertain what heating process is suitable for use with the identified aerosol generating material **132**. Once age verification is satisfied or deemed unnecessary, the device **110** may offer the user one of a number of heating processes for use with the aerosol generating material **132** to improve user experience. This may be part of the updating of the activation state of the aerosol provision device **110**.
[0029] In a specific example, the control circuitry **120** is arranged to update an activation state of the aerosol provision device **110** to a non-operating state in response to receipt of a signal from the detector **130** associated with the first property having or exceeding a predetermined value. For example, where the aerosol generating material **132** contains a nicotine value above negligible or zero, the system **100** is prevented from providing an aerosol to the user until age verification is satisfied. In this way, the user is protected from use of unsuitable aerosol generating material. The user may interact with the age verification means **140** to provide an age, where the device **110** may be allowed then to provide an aerosol.
[0030] In the above specific example, the control circuitry **120** is arranged to activate the age verification means **140** in response to receipt of a signal from the detector **130** associated with the first property (e.g. nicotine content) having or exceeding a predetermined value.
[0031] In a specific example, the control circuitry **120** is arranged to update an activation state of the aerosol provision device **110** to an operating state in response to receipt of a signal from the detector **130** associated with the first property being less than a predetermined value. For example, where the aerosol generating material **132** contains a nicotine value that is negligible or zero, the system **100** is allowed to provide an aerosol to the user without satisfying age verification. This may also be applied to other suitable age-restricted compounds.
[0032] The age verification means **140** may comprise at least one of: a face scanner; a fingerprint scanner; an iris scanner; a microphone; a camera; a mobile device; and, a communication module for communicating with a database.
[0033] In the example of a camera, the age verification means **140** may factor in aspects such as lines in the face, bags under the eyes, and prevalence of e.g. in males, facial hair to provide an age verification means. Such factors can be used to indicate a probable age range of the potential user of the device **110**. The age verification means **140** may perform an analysis of the data taken of the user, and provide a signal to the control circuitry **120** indicating whether the user is of a likely suitable or likely unsuitable age for use of the device. Alternatively or additionally, the camera may be used to inspect a users identification, such as a passport to the like, to detect an age of the user. If there is an image on the identification, this can be compared against the user's face using the camera.
[0034] In the example of FIG. **1**, the age verification means **140** is not integral with the aerosol provision device **110**. In the example of FIG. **1**, the detector **130** is not integral with the aerosol provision device **110**.

[0035] Referring now to FIG. **2**, there is a shown a similar system **200** to the system **100** of FIG. **1**. Similar features, to those features used in FIG. **1**, are shown with the reference numerals increased by 100. For example, the system **100** of FIG. **1** is similar to the system **200** of FIG. **2**. Similar or identical features may not be discussed for conciseness.
[0036] The system **200** of FIG. **2** has aerosol provision device **210** comprising the control circuitry **220**, detector **230**, and aerosol generating material **232**. The age verification means **240** is located outside of the aerosol provision device **210**.
[0037] In the example shown in FIG. **2**, the aerosol generating material **232** is shown in-use as located inside the aerosol provision device **210** forming a portion of the aerosol provision system **200**. In this example, the aerosol generating material **232** may be inserted into the housing of the aerosol provision device **210**. In this way, when located inside the aerosol provision device **210**, the aerosol provision device **210** may identify, via detector **230**, the aerosol generating material **232**. The aerosol generating material **232** may be in the form of a pod or a cartridge containing aerosol generating material which is inserted into the aerosol provision device **210** and then discarded or refilled after use.
[0038] The age verification means **240** may be separate to the aerosol provision device **210** and used separately, while communicatively coupled to the control circuitry **220**. The control circuitry **220** may provide a notification to alert the user that age verification is required and to conduct such through the age verification means **240**. The age verification means **240** may be a mobile phone or the like.
[0039] The age verification means **240** receives a signal from the control circuitry **220** corresponding to a request to operate and perform age verification on the user. The age verification means **240** may detect a property associated with a potential user and provide age verification. The age verification means **240** then provides a signal back to the control circuitry **220** corresponding to whether or not the potential user has satisfied age verification. Where the user satisfies age verification, the control circuitry **220** provides a signal to heaters or the like to function and provide an aerosol from the aerosol generating material **232** in the aerosol provision device **210**.
[0040] In the event that the age verification means **240** detects the user is in an age range (or of an age) suitable for using the system **200** in all operational modes (such as heating age restricted aerosol generating material), the control circuitry **220** may allow any such activation of the device **210**. In the event that the age verification means **240** detects the user is not in an age range (or of an age) suitable for use with the aerosol generating material **232** the control circuitry **220** may signal to the user that the aerosol generating material **232** is to be removed from the aerosol provision device **210**. Notifications to the user may occur via a screen on the aerosol provision device **210** or any other suitable means such as a display on the age verification means **240**.
[0041] This arrangement allows suitable handling of a device that is used by multiple users of different age. The users of suitable age are allowed to access suitable operational states for respective aerosol generating materials, while users of un-suitable age are prevented from doing so. This ensures that suitability of use is prioritised alongside user safety.
[0042] In an example, the control circuitry **220** is arranged to update an activation state of the aerosol provision device **210** in response to receipt of an access signal or deny signal from the age verification means **240**. In response to receipt of an access signal, the control circuitry **220** may update the device **210** to a full access mode where all functions are allowed for the aerosol generating material **232**. In response to receipt of a deny signal, the control circuitry **220** may update the device **210** to a no access mode, where all functions are denied or only partial functionality is provided for the aerosol generating material **232**.
[0043] In an example, the control circuitry **220** is arranged to provide a notification to a user in response to receipt of a deny signal from the age verification means **240**. The notification may be in the form of an audible signal, a visual signal, a haptic signal, a textual message or the like displayed on a screen or display or the like. Any suitable notification method may be used,

including sending a signal to a personal device of the user, such as a smart device or the like. The age verification means **240** may be a smart device. The notification may be provided from the age verification means **240**.

[0044] Referring now to FIG. **3**, there is a shown a similar system **300** to the system **200** of FIG. **2**. Similar features, to those features used in FIG. **2**, are shown with the reference numerals increased by 100. For example, the system **200** of FIG. **2** is similar to the system **300** of FIG. **3**. Similar or identical features may not be discussed for conciseness.

[0045] The device **310** of FIG. **3** is shown containing the control circuitry **320**, detector **330**, aerosol generating material **332** and an age verification means **340**. The system **300** is therefore more compact than the system **200** of FIG. **2**. There is no need for a separate age verification means in the arrangement of FIG. **3**. This may improve the user experience as the user need not carry multiple items for accessing the device **310**.

[0046] Referring now to FIG. **4**, there is a shown a similar system **400** to the system **300** of FIG. **3**. Similar features, to those features used in FIG. **3**, are shown with the reference numerals increased by 100. For example, the system **300** of FIG. **3** is similar to the system **400** of FIG. **4**. Similar or identical features may not be discussed for conciseness.

[0047] The device **410** of FIG. **4** is shown containing the control circuitry **420**, detector **430**, aerosol generating material **432** and an age verification means **440**. The age verification means **440** comprises an additional element **442**. The element **442** may be a communication element to allow the age verification means **440** to communicate with a remote database. The element **442** may be a memory storing an on board database. The element **442** may be both a communication element and a memory.

[0048] In the above example, the age verification means **440** receives signals from the control circuitry **420** and provides a notification to the user to perform age verification. The age verification means obtains a signal from the user and provides to the element **442**.

[0049] The element **442** may be a database stored in a memory on board the device **400**. The signal from the from age verification means **440** is compared to the on board database. If the signal is associated with a user of suitable age, the device **410** is updated into an operational state. This on board database arrangement may be advantageous as the device **400** need not have a communications element in the device **400** to communicate with a remotely held database, and the device **400** need not be connected to a communications network to access a remotely held database prior to each use session. This may allow use of the device **410** in areas without connectivity. This may also provide a faster response than via communicating with a remote database.

[0050] In a different example, the database of user age is held remotely, and the age verification means **440** has a communication module **442** to contact the database. The communication module **442** may contact the database with a request for confirming user age. The communication module **442**, and therefore the age estimation module **440**, is then provided with an age of the potential user, which is relayed to the control circuitry **420**. This arrangement may be advantageous as the device **410** need not include a memory element for carrying the database and the database can be remotely updated ensuring the device **410** need not have the on board database regularly updated. In this way up-to-date age data can be provided to all devices **410** as soon as the data is uploaded to the central database. In this way, all users can be provided with the updates without each needing to update their own device **410**. The remote database may be a governmental database or the like for linking a property of a user to a known age for that user, or for verifying the identification of a user where the identification contains an indication of age for that user.

[0051] The present invention involves changing an activation sate of the aerosol provision device. In an “operating state”, elements of the aerosol provision device **410** used to generate an aerosol (such as an atomiser, heater or the like) may be activated. The specific activation of the device **410** may require an additional input which may be inhalation on the device **410**, pressing a button on the device **410** or the like. Alternatively, the device **410** may automatically generate aerosol by a heater in response to receiving a signal associated with a valid potential user. The control circuitry **420** may receive such a signal from the age verification means **440** and send a signal to the heater arrangement or the like to provide an aerosol from an aerosol generating material **432** that may contained within, or separate to, the aerosol provision device **410**.

[0052] FIG. **5** shows a method **500** of use of an aerosol provision device. In the method **500**, the device may start in a default state **502**, which may be a non-operating state such that when age restricted aerosol generating material is provided to the device, users cannot use the device without age verification or the detection of no age-restricted aerosol generating material being present. Alternatively, the default operating state may be a restricted operating state where only partial operation of the device is possible. Alternatively, the device may be an operating mode and is prevented from operating in response to detection of age-restricted aerosol generating material.

[0053] When a user intends to use the aerosol provision device, the user inputs the aerosol generating material to the device. The device detects a first property of the aerosol generating material **404**. The device detects the first property using a detector (which may contain a number of individual sensors/detectors). The detector may detect the property from a barcode, QR code, label or the like, or may detect the chemical composition of the aerosol generating material directly.

[0054] The detector sends a signal accordingly to the control circuitry **506**. An assessment is made whether age verification is required or not—this may occur at the detector or the control circuitry. If age verification is not required, the activation state of the aerosol provision device may be updated **508**, if it is, the age verification means may be activated **508**.

[0055] This method provides a user-friendly age verification process that does not impede overly use for non age-restricted aerosol generating materials. The method offers a balance between overly strict and overly lenient access protection for the device and aerosol generating material.

[0056] The updating of the activation state may involve updating, by the control circuitry, an activation state of the aerosol provision device to an operating state in response to receiving a signal from the detector associated with the first property being less than a predetermined value. The updating of the activation state may involve updating, by the control circuitry, an activation state of the aerosol provision device to a non-operating state in response to receiving a signal from the detector associated with the first property having or exceeding a predetermined value.

[0057] The term “in response to” is used herein to indicate a second event (such as a signal or change of state of an aerosol provision device) that occurs subsequent to a first event. The second event may occur at a later time, after a predetermined time, or immediately after the first event.

[0058] The device and system herein are described as comprising several components that enable several advantages. The components may be disclosed as on-board the device or within the system. The components may be distributed and therefore not necessarily be located on-board the device. The functionality of the device can be provided by communicatively connected components, and such communication may be wireless, enabling such distribution. At which point it is reasonable to foresee that a distributed array of components will operate in the manner of the devices and systems disclosed herein. Components of the device or system may be contained in a further device such as a smartphone, computer, or remote server or the like.

[0059] The method and device disclosed herein enable protection over the use of the device without requiring an arduous authorisation process—the process is only as arduous as is required considering the content of the aerosol generating material to be used. This improves the user experience of the device and the safety of general use of the device.

[0060] In a particular example, the device disclosed herein may operate with a flavour pod which is replaceable in the device—this may be referred to as a consumable. The flavour may be any of tobacco and glycol and may include extracts (e.g., licorice, hydrangea, Japanese white bark magnolia leaf, chamomile, fenugreek, clove, menthol, Japanese mint, aniseed, cinnamon, herb, wintergreen, cherry, berry, peach, apple, Drambuie, bourbon, scotch, whiskey, spearmint, peppermint, lavender, cardamon, celery, cascarilla, nutmeg, sandalwood, bergamot, geranium, honey essence, rose oil, vanilla, lemon oil, orange oil, cassia, caraway, cognac, jasmine, ylang-ylang, sage, fennel, piment, ginger, anise, coriander, coffee, or a mint oil from any species of the genus Mentha), flavour enhancers, bitterness receptor site blockers, sensorial receptor site activators or stimulators, sugars and/or sugar substitutes (e.g., sucralose, acesulfame potassium, aspartame, saccharine, cyclamates, lactose, sucrose, glucose, fructose, sorbitol, or mannitol), and other additives such as charcoal, chlorophyll, minerals, botanicals, or breath freshening agents. They may be imitation, synthetic or natural ingredients or blends thereof.

[0061] When combined with an aerosol generating medium, the aerosol provision device as disclosed herein may be referred to as an aerosol provision system.

[0062] Thus there has been described an aerosol-generating system, for providing an aerosol for inhalation by a user, comprising: an aerosol provision device comprising control circuitry for controlling an activation state of the aerosol provision device; a detector arranged to detect a first property associated with aerosol generating material, in use arranged in the aerosol provision device, and provide a signal to the control circuitry; and, age verification means for verifying an age of a user, wherein the control circuitry is arranged to perform at least one of i) update an activation state of the aerosol provision device, and ii) activate the age verification means in response to receiving a signal from the detector.

[0063] The aerosol provision system may be used in a tobacco industry product, for example a non-combustible aerosol provision system.

[0064] In one embodiment, the tobacco industry product comprises one or more components of a non-combustible aerosol provision system, such as a heater and an aerosolizable substrate.

[0065] In one embodiment, the aerosol provision system is an electronic cigarette also known as a vaping device.

[0066] In one embodiment the electronic cigarette comprises a heater, a power supply capable of supplying power to the heater, an aerosolizable substrate such as a liquid or gel, a housing and optionally a mouthpiece.

[0067] In one embodiment the aerosolizable substrate is contained in or on a substrate container. In one embodiment the substrate container is combined with or comprises the heater.

[0068] In one embodiment, the tobacco industry product is a heating product which releases one or more compounds by heating, but not burning, a substrate material. The substrate material is an aerosolizable material which may be for example tobacco or other non-tobacco products, which may or may not contain nicotine. In one embodiment, the heating device product is a tobacco heating product.

[0069] In one embodiment, the heating product is an electronic device.

[0070] In one embodiment, the tobacco heating product comprises a heater, a power supply capable of supplying power to the heater, an aerosolizable substrate such as a solid or gel material.

[0071] In one embodiment the heating product is a non-electronic article.

[0072] In one embodiment the heating product comprises an aerosolizable substrate such as a solid or gel material, and a heat source which is capable of supplying heat energy to the aerosolizable substrate without any electronic means, such as by burning a combustion material, such as charcoal.

[0073] In one embodiment the heating product also comprises a filter capable of filtering the aerosol generated by heating the aerosolizable substrate.

[0074] In some embodiments the aerosolizable substrate material may comprise an aerosol or aerosol generating agent or a humectant, such as glycerol, propylene glycol, triacetin or diethylene glycol.

[0075] In one embodiment, the tobacco industry product is a hybrid system to generate aerosol by heating, but not burning, a combination of substrate materials. The substrate materials may comprise for example solid, liquid or gel which may or may not contain nicotine. In one embodiment, the hybrid system comprises a liquid or gel substrate and a solid substrate. The solid substrate may be for example tobacco or other non-tobacco products, which may or may not contain nicotine. In one embodiment, the hybrid system comprises a liquid or gel substrate and tobacco.

[0076] In order to address various issues and advance the art, the entirety of this disclosure shows by way of illustration various embodiments in which the claimed invention(s) may be practiced and provide for a superior electronic aerosol provision system. The advantages and features of the disclosure are of a representative sample of embodiments only, and are not exhaustive and/or exclusive. They are presented only to assist in understanding and teach the claimed features. It is to be understood that advantages, embodiments, examples, functions, features, structures, and/or other aspects of the disclosure are not to be considered limitations on the disclosure as defined by the claims or limitations on equivalents to the claims, and that other embodiments may be utilised and modifications may be made without departing from the scope and/or spirit of the disclosure. Various embodiments may suitably comprise, consist of, or consist essentially of, various combinations of the disclosed elements, components, features, parts, steps, means, etc. In addition, the disclosure includes other inventions not presently claimed, but which may be claimed in future.

Claims

1. An aerosol-generating system, for providing an aerosol for inhalation by a user, comprising: an aerosol provision device comprising control circuitry for controlling an activation state of the aerosol provision device; a detector arranged to detect a first property associated with aerosol generating material, in use arranged in the aerosol provision device, and provide a signal to the control circuitry; and, age verification means for verifying an age of a user, wherein the control circuitry is arranged to perform at least one of i) update an activation state of the aerosol provision device, and ii) activate the age verification means in response to receiving a signal from the detector.

2. An aerosol-generating system according to claim 1, wherein the first property associated with the aerosol generating material is at least one of: chemical composition; amount; date of production; and, batch number.

3. An aerosol-generating system according to claim 1, wherein the control circuitry is arranged to update an activation state of the aerosol provision device to a non-operating state in response to receipt of a signal from the detector associated with the first property having or exceeding a predetermined value.

4. An aerosol-generating system according to claim 1, wherein the control circuitry is arranged to update an activation state of the aerosol provision device to an operating state in response to receipt of a signal from the detector associated with the first property being less than a predetermined value.

5. An aerosol-generating system according to claim 1, wherein the control circuitry is arranged to activate the age verification means in response to receipt of a signal from the detector associated with the first property having or exceeding a predetermined value.

6. An aerosol-generating system according to claim 1, wherein the age verification means comprises at least one of: a face scanner; a fingerprint scanner; an iris scanner; a microphone; a camera; a mobile device; and, a communication module for communicating with a database.

7. An aerosol-generating system according to claim 1, wherein the age verification means is not integral with the aerosol provision device.

8. An aerosol-generating system according to claim 1, wherein the detector is not integral with the aerosol provision device.

9. An aerosol provision device, for providing an aerosol for inhalation by a user, comprising: control circuitry for controlling an activation state of the aerosol provision device; a detector arranged to detect a first property associated with aerosol generating material, in use arranged in the aerosol provision device, and provide a signal to the control circuitry; and, age verification means for verifying an age of a user, wherein the control circuitry is arranged to perform at least one of i) update an activation state of the aerosol provision device, and ii) activate the age verification means in response to receiving a signal from the detector.

10. An aerosol provision device according to claim 9, wherein the first property associated with the aerosol generating material is at least one of: chemical composition; amount; date of production; and, batch number.

11. An aerosol provision device according to claim 9, wherein the control circuitry is arranged to update an activation state of the aerosol provision device to a non-operating state in response to receipt of a signal from the detector associated with the first property having or exceeding a predetermined value.

12. An aerosol provision device according to claim 9, wherein the control circuitry is arranged to update an activation state of the aerosol provision device to an operating state in response to receipt of a signal from the detector associated with the first property being less than a predetermined value.

13. An aerosol provision device according to claim 9, wherein the control circuitry is arranged to activate the age verification means in response to receipt of a signal from the detector associated with the first property having or exceeding a predetermined value.

14. An aerosol provision device according to claim 9, wherein the age verification means comprises at least one of: a face scanner; a fingerprint scanner; an iris scanner; a microphone; a camera; a mobile device; and, a communication module for communicating with a database.

15. A method of providing an aerosol for inhalation by a user, the method comprising: detecting, by a detector, a first property associated with aerosol generating material arranged in an aerosol provision device; providing, by the detector, a signal to control circuitry of the aerosol provision device; in response to receiving, at the control circuitry, a signal from the detector, performing at least one of: updating, by the control circuitry, an activation state of the aerosol provision device; and activating, by the control circuitry, age verification means.

16. The method of claim 15, wherein detecting, by a detector, a first property associated with aerosol generating material comprises detecting at least one of: chemical composition; amount; date of production; and, batch number associated with the aerosol generating material.

17. The method of claim 15, wherein updating, by the control circuitry, an activation state of the aerosol provision device comprises updating, by the control circuitry, an activation state of the aerosol provision device to a non-operating state in response to receiving a signal from the detector associated with the first property having or exceeding a predetermined value.

18. The method of claim 15, wherein updating, by the control circuitry, an activation state of the aerosol provision device comprises updating, by the control circuitry, an activation state of the aerosol provision device to an operating state in response to receiving a signal from the detector associated with the first property being less than a predetermined value.

19. The method of claim 15, wherein activating, by the control circuitry, age verification means comprises activating the age verification means in response to receiving of a signal from the detector associated with the first property having or exceeding a predetermined value

20. Aerosol-generating means, for providing an aerosol for inhalation by a user, comprising: an aerosol provision device comprising control means for controlling an activation state of the aerosol provision device; detecting means arranged to detect a first property associated with aerosol generating material, in use arranged in the aerosol provision device, and provide a signal to the control means; and, age verification means for verifying an age of a user, wherein the control means is arranged to perform at least one of i) update an activation state of the aerosol provision device, and ii) activate the age verification means in response to receiving a signal from the detecting means.